Disparity Analyses Are Robust To Ancestral State Estimation Uncertainty

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Abstract

Ancestral state estimation is a widely used method to infer unobserved traits across a phylogeny, extending analyses beyond incomplete fossil records. In studies of morphological disparity ancestral state estimation can account for ghost lineages, yet estimates are often confounded with high uncertainty. How best to deal with this uncertainty remains unclear. Using simulations, we compared pre- and post-ordination estimation methods, as well as both point estimate and probabilistic approaches for treating uncertainty, under continuous and discrete trait models and varying fossil sampling levels. The relative disparity error was calculated from the difference between our estimated disparity values and the true disparity values from the original simulations. We then extended this framework to evaluate the recovery of changes in disparity across time by simulating mass extinction events with varying intensity and selectivity. We find that pre-ordination probabilistic estimations outperform post-ordination and point estimation methods and that omission of ancestral state estimations leads to underestimates of disparity. As expected, higher fossil sampling improves estimation accuracy, although recovered disparity accuracy was robust even at low fossil sampling. Similar patterns were observed when recovering changes in disparity across mass extinction events but declined significantly when fossil sampling was low, highlighting that fossil inclusion is essential when assessing disparity in deep time. These results demonstrate that explicitly propagating uncertainty, rather than collapsing states into point estimates, is more likely to recover accurate patterns in disparity, reinforcing the need to move beyond point estimates when working with the incomplete fossil record.

Keywords: ancestral state estimation; morphological disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

Simulations were carried out using `treats` (Guillerme 2024). Phylogenies were simulated using a modified version of the birth-death process algorithm. The process begins with a single lineage and iteratively selects lineages to grow the tree forward in time. At each step, all living branches are extended by a branch length increment, drawn from an exponential distribution based on the total number of lineages and the combined speciation and extinction rates. Selected lineages then undergo either speciation or extinction based on a probabilistic decision determined by the specified rates. Phylogenies were simulated with birth-death parameters as speciation (λ) = 1.0, and extinction (μ) across a range between 0.9-0.5. Fixing $\lambda = 1.0$ provides a convenient relative time-scale, since only the ratio of extinction to speciation affects the shape and balance of simulated trees. The chosen parameters correspond to a moderate-to-high turnover regime, consistent with the upper-empirical estimates from both fossil- and molecular-based diversification studies of vertebrate groups. This level of extinction produces realistic evolutionary trees with substantial lineage pruning and extinct diversity, while maintaining a positive net diversification rate (λ - μ = 0.3). However, these parameters were not fixed, since they can vary if a tree fails to be generated with given stop rule and parameters. A histogram of the distribution of speciation/extinction ratios is in the supplementary. 100 trees were simulated for each of three different stop rules: 50, 100 and 150 living taxa.

2.1.2 Trait simulation

Across each phylogeny, we simulated discrete and continuous trait data separately. Discrete trait data were simulated across three rate categories, representing fast, medium and slow transition rates (table 1). These rates were assigned by generating equal-rate (ER) transition probability matrices, i.e., $P_{ij} = P_{ji}$. For each category, 85 binary characters and 15 three-state characters were simulated, following the character coding ratios often found in discrete datasets used for disparity analyses (Casali et al. 2023). Discrete trait evolution was simulated using a discrete-time Markov process, where the probability of state transition along each branch was proportional to

the product of the transition probability and branch length.

Continuous trait data were simulated under variations of Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models (table 1). For BM, the trait data were simulated as both stationary and with a moving trend. For OU, the data were simulated as strong, weak and with differing optimum and root states. Following Gearty et al. (2024), the alpha (α) parameter was set based on phylogenetic half life ($\frac{ln(2)}{\alpha}$) to account for differing tree heights across simulations (Cooper et al. 2016). For the strong and weak OU models α was set as 10% and 50% of the tree height respectively. Continuous trait data were simulated across 100 characters for each model across each phylogeny. Each character was simulated as independent.

2.2 Fossil sampling

In order to simulate different levels of fossil sampling, each tree was split into chronological 'stages' to emulate the stratigraphic nature of the fossil record. Fossil tips were sampled uniformly at random from all extinct lineages within each chronological stage. Fossil sampling was carried out at five levels, ω_0, ω_5, ω_{15}, ω_{50} and ω_{100} (where $\omega_P = P\%$ fossils sampled), across each simulated phylogeny. The corresponding trait matrices were subset to match sampled taxa, with ancestral node states excluded.

2.3 Ancestral state estimation

Ancestral state estimation was carried out on the down-sampled matrices using the `multi.ace` function in `dispRity`, which is a wrapper for `ace` in the `ape` package (Paradis and Schliep 2019). We used four combinations of two approaches to ancestral state estimation to evaluate each method's accuracy in recovering the true disparity signal.

2.3.1 Pre-ordination estimation

Firstly, we compared pre- versus post-ordination estimation methods. Pre-ordination estimation methods of discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a Maximum-Likelihood derived estimate of the Mk transition matrix and the tree topology given (Pagel 1999; Yang 2006). Given that the transition rates were set *a priori* during simulations, the equal-rates (ER) model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating 2023).

Pre-ordination estimation of continuous characters models evolution under a Brownian motion model, whereby ancestral state values lie within a multivariate normal distribution that is estimated given the observed tip data and the shared evolutionary history between taxa, i.e., the tree topology (Schluter et al. 1997; Rohlf 2001). Confidence intervals were calculated from standard errors derived from the Hessian matrix at the maximum likelihood estimate. The width of confidence intervals reflects the uncertainty in continuous ancestral estimates, which typically increases with node depth and decreases with the number of informative descendant lineages. For a recent review of the methods described here see Revell (2025).

2.3.2 Post-ordination estimation

Post-ordination estimation involves the estimation of a node's position within a multidimensional ordination matrix (or trait space). The generation of an ordination matrix differed between discrete and continuous characters. For discrete characters, distance matrices were computed using the 'MORD' metric (Lloyd 2016) and ordinated using principal coordinates analyses (PCoA) (`cmdscale()` function). Continuous data were ordinated using principal components analysis (PCA) (`prcomp()` function). For both the discrete- and continuous- character derived trait spaces, the ancestral states across all PC axes were then estimated under a Brownian motion model, as detailed in section 2.3.1.

2.3.3 Uncertainty treatment

For each estimation method, we also treated uncertainty in estimates with both deterministic and probabilistic approaches. The deterministic approach involves collapsing the likelihood outputs into point estimates. For discrete characters, this involved a strict majority rule whereby the state with the highest scaled likelihood was selected at each node for each character. In scenarios where state likelihoods are equally tied, one state is collapsed at random. We also employed a relative majority rule, whereby any state at a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states is collapsed, otherwise states are left as uncertain. For continuous characters, the deterministic approach used the median of the estimated ancestral confidence interval.

The probabilistic approach differs in that uncertainty in estimates are explicitly accounted for through the replication of matrices. For instance, discrete states at each node for each character were 'sampled' according to their scaled likelihoods across 100 matrix replications. In a scenario

where there is uncertainty in a binary character for a specific node, e.g., 0.55 for state 0 and 0.45 for state 1, this probabilistic approach would have a 55% probability of sampling state 0 in each matrix for the respective node. Continuous characters were treated similarly, with 100 matrices replicated and ancestral state estimates for each node and character sampled across a normal distribution within their respective estimated confidence interval. Both procedures are implemented in the `dispRity` package using the `sample = 100` argument in the `multi.ace` function (Guillerme 2018).

2.4 Disparity metrics

To ensure fair comparison between different fossil sampling levels only the nodes present across all fossil sampling levels were retained from the ancestral state estimation outputs for each phylogeny; therefore, comparisons in disparity calculations between fossil sampling levels differed solely on the difference in number of fossil tips and the estimations of node states. Discrete and continuous pre-ordination ancestral state estimations were ordinated in the same manner as described in section 2.3.2.

When assessing overall disparity, we wanted to understand how well ancestral state estimation can recover true patterns in size and density in trait space (Guillerme et al. 2020). Since we were assessing overall disparity, position was not a metric that can be discerned due to the centring of ordinated matrices. Size-based metrics describe the spread of points in the trait space (Guillerme et al. 2020). To explore how well the size of the trait space was recovered, we calculated the sum of quantiles. The sum of quantiles calculates the range on each PC axis for the 95% of the data and then sums these ranges across all axes. This metric has similar properties to the often used sum of ranges metric by describing the total trait space exploited (Foote 1992), but with less sensitivity to outliers. For density, we measured the sum of variances (Ciampaglio et al. 2001; Wills 2001) and the average nearest neighbour distance (Foote 1992). The sum of variances is the most widely used metric in disparity analyses. Disparity values were calculated across each estimation and the 'true' simulation so that a relative disparity error could be calculated.

$$\text{Relative disparity error} = \frac{\text{Estimated disparity} - \text{True disparity}}{\text{True disparity}} \tag{1}$$

Disparity calculations were carried out across three treatments of the ordinated matrices. Firstly, the untreated ordination output was used. Secondly, we min-max standardised the PC

axes so that the absolute magnitude in PC scores did not influence the disparity metric scores. Thirdly, we only used the minimum number of dimensions that were contained across all fossil sampling levels for that specific phylogeny. For the simulations of 50 living taxa trees, this would be the first 48 dimensions, in the 100 living taxa it would be 98 dimensions etc.

2.5 Mass extinction simulations

Having established the accuracy of different ancestral state estimation methods in recovering overall trait space patterns, we next tested how well these methods capture changes in disparity across time during simulated mass extinction events. Following the framework above, 100 trees were simulated for each tree size of 50, 100 and 150 living taxa with a mass extinction event. We simulated both random, where taxa were eliminated at non-selectively, and selective extinctions, where taxa were more prone to extinction based on their trait values. Extinction severity was set with a uniform distribution between 75–95%, the range seen between K-Pg and the Permian extinctions. For selective extinctions,

With the new set of trees (and trait data for selective extinctions), the framework from section 2.1.2 to 2.3.3 was repeated. Rather than evaluating the overall disparity estimate, we assessed how well the methods could recover the relative disparity change across the extinction event. Two bins that were equidistant based off the total tree height divided by six were sampled either side of the extinction event. Disparity was measured from the taxa contained within each of these two bins to calculate a relative disparity change across the extinction event.

$$\text{Relative change in disparity} = \frac{\text{Disparity post-extinction} - \text{Disparity pre-extinction}}{\text{Disparity pre-extinction}} \tag{2}$$

The sum of quantiles, sum of variances and average nearest neighbour distance were calculated again, as well as the average displacement (Guillerme et al. 2020) as a change in position metric.

3 Results

3.1 Overall disparity error

3.1.1 Discrete

3.2 Overall Performance of ASE Methods

4 Discussion

5 Conclusions

Acknowledgments

Author Contributions

Data Availability

Competing Interests

Figures

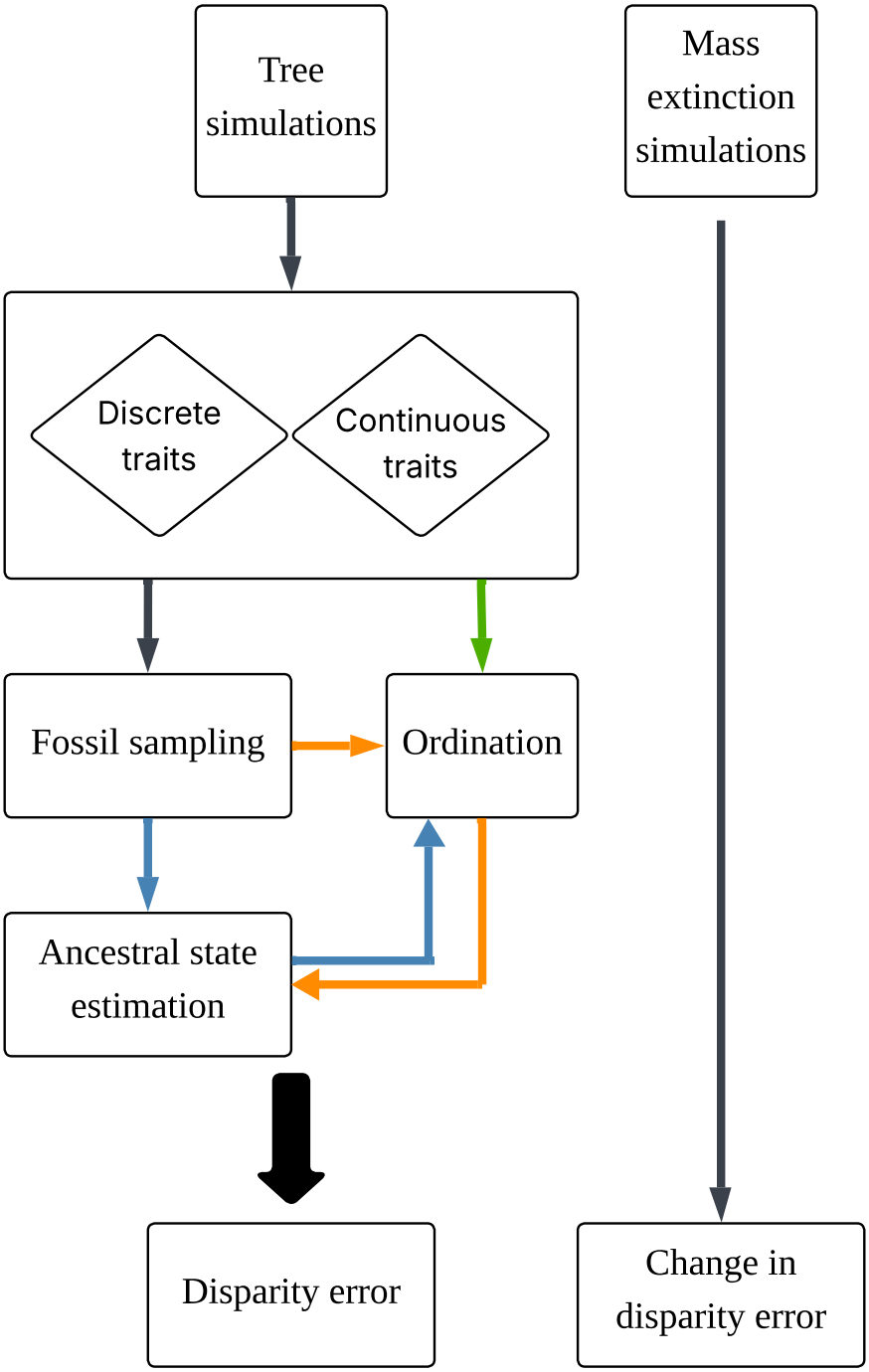

Figure 1: **Simulation workflow**
Green arrow = 'true disparity'; blue arrow = pre-ordination estimation; orange arrow = post-ordination estimation.

Tables

Table 1: Trait simulation models and parameters used across each phylogeny. For continuous characters, the root state was always set as 0. P_{ij} = transition probability; σ = variance; α = rate of adaptation toward the optimum; θ = optimum.

Type of data	**Model**	**Parameters**
Discrete	Fast	$P_{ij} = 0.9$
	Medium	$P_{ij} = 0.1$
	Slow	$P_{ij} = 0.01$
Continuous	BM	$\sigma = 0.25$
	BM (trend)	$\sigma = 0.25,\ \text{trend} = 0.3$
	OU (strong)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 10}}$
	OU (weak)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 5}}$
	OU (shift optimum)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 5}},\ \theta = 2$

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